

Practical No 6

Aim: Develop a program to show polymorphism by implementing method overloading and method overriding concept in java.

Source Code:

```
class Shape {
    void area(){
        System.out.println("Calculating area.....");
    }
    void area(int side){
        System.out.println("Area of Square:"+(side*side));
    }
    void area(int length, int width){
        System.out.println("Area of Rectangle:"+(length*width));
    }
    void display(){
        System.out.println("This is a Shape.");
    }
}
class Circle extends Shape{
    void display(){
        System.out.println("This is a Circle.");
    }
}
class polymorphismDemo{
    public static void main(String[] args) {
        Shape s= new Shape();
        s.area();
        s.area(5);
        s.area(4,6);
        Shape ref;
        ref=new Shape();
        ref.display();
        ref=new Circle();
        ref.display();
    }
}
```

Output:

```
Calculating area.....
Area of Square:25
Area of Rectangle:24
This is a Shape.
This is a Circle.
```